

1. Administrative & Study Identification

Study Title: A Study to Compare the Efficacy and Safety of Oral Azacitidine Plus Best Supportive Care (BSC) Versus Placebo Plus BSC in Participants With International Prognostic Scoring System Revised (IPSS-R) Low- or Intermediate-risk Myelodysplastic Syndrome (MDS) **Official Title:** A Phase 2/3, Multicenter, Randomized, Dose Optimization (Part I), Double-blind (Part II) Study to Compare the Efficacy and Safety of Oral Azacitidine (Oral-Aza, ONUREG®) plus Best Supportive Care (BSC) versus Placebo plus BSC in Participants with IPSS-R Low- or Intermediate-risk Myelodysplastic Syndrome (MDS) **Short Title/ Acronym:** ASTREON **Protocol Number:** CA055-026 **NCT Number:** NCT05469737 **Posting ID:** 72607804122006579 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Oral Azacitidine (Oral-Aza, ONUREG®) **Phase of Development:** Phase 2, Phase 3 (PHASE2, PHASE3) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Summary/Primary Objective: The purpose of this study is to evaluate the safety and efficacy of oral azacitidine in participants with low to intermediate International Prognostic Scoring System Revised (IPSS-R) myelodysplastic syndrome (MDS). It includes evaluating the recommended Phase 3 dose (Part I) and comparing complete remission (CR) rates of oral azacitidine plus best supportive care (BSC) versus placebo plus BSC (Part II). **Secondary Objectives:** To evaluate the achievement of overall response (OR) rate, 84-day packed red blood cells transfusion independence (pRBC-TI), pRBC-TI duration, platelet-TI, overall survival (OS), event-free survival, and time to acute myeloid leukemia (AML) progression.

2.2 Background & Rationale To address the unmet medical need for new treatment options in lower-risk MDS to manage cytopenias and prevent progression to higher-risk MDS and acute myeloid leukemia (AML). Early trials established the clinical benefit of oral azacitidine; this trial aims to evaluate an optimized 14-day dosing regimen to minimize hematologic and gastrointestinal toxicities while maintaining durable clinical efficacy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (in Part II) **Blinding:** Open-label (Part I for dose optimization), Double-blind (Part II) **Randomization:** Randomized (1:1 ratio)

3.2 Planned Enrollment & Duration Targeted Enrollment: 230 participants (~42 in Phase 2, ~188 in Phase 3) **Number of Sites:** Multicenter, global **Estimated Study Start:** December 2022 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Participant has a documented diagnosis of MDS according to WHO 2016 classification that meets International Prognostic Scoring System Revised (IPSS-R) classification of low- or intermediate-risk disease (IPSS-R score between 1.5 and 4.5). MDS diagnosis, WHO classification, and IPSS-R risk classification will be prospectively determined by independent central pathology and cytogenetics review, and applicable central laboratory results. 3. Participant must have an Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, or 2. 4. Presence of $\geq 1\%$ cytopenia (anemia, thrombocytopenia, or neutropenia).

4.2 Exclusion Criteria 1. Participants with prior malignancies must have an expected median life expectancy of at least 12 months at the time of inclusion and no active treatment of any sort for at least 24 weeks prior to randomization (including but not limited to immunotherapy or targeted therapy). 2. Hypoplastic Myelodysplastic Syndrome (MDS) with a marrow cellularity of $\leq 10\%$. 3. Participants diagnosed with MDS with excess blasts-2 (MDS-EB2). 4. Prior treatment with azacitidine (any formulation), decitabine, or other hypomethylating agent. 5. Absolute neutrophil count $< 0.5 \times 10^9/L$ within a week of randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name / Preferred Name: azacitidine **Trade Names:** Vidaza, Onureg **ATC Code:** L01BC07 **EMA URLs:** * https://www.ema.europa.eu/en/documents/product-information/onureg-epar-product-information_en.pdf * https://www.ema.europa.eu/en/documents/product-information/vidaza-epar-product-information_en.pdf **Dosage/Frequency:** Oral Azacitidine 200 mg or 300 mg once daily (QD) for 14 days per 28-day cycle. **Route of Administration:** Oral **Duration of Treatment:** 6 cycles for primary endpoint assessment; patients who benefit may continue in an extension phase.

5.2 Concomitant & Prohibited Therapy Permitted: Best Supportive Care (BSC) including transfusions and standard cytopenia management. **Prohibited:** Other hypomethylating agents, decitabine, or concurrent investigational cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, IPSS-R Scoring, Central Pathology/Cytogenetics Review, Baseline Labs
Baseline	Day 0	Randomization, First Dose (Cycle 1, Day 1)
Interim	Every 28 Days	Day 1 of each cycle: Safety Labs, AE monitoring, Efficacy Assessment (CBC counts for TI)
End of Study	Post-Cycle 6	Disease assessment (CR/ORR), transition to extension phase if benefiting, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Number of participants with Adverse Events (AEs) evaluated using the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) criteria v. 5.0: Phase 2. 2. Number of participants who achieved complete remission (CR) per International Working Group (IWG) 2006 criteria within 6 cycles: Phase 2 and 3.

7.2 Secondary Outcomes 1. Number of participants who achieved Overall Response (OR) per IWG 2006 criteria within 6 cycles. 2. Number of participants who achieved 84-day packed red blood cells transfusion independence (pRBC-TI). 3. pRBC-TI duration. *(Additional exploratory outcomes include platelet-TI sustained for 84 days, OS, event-free survival, and time to AML progression).*

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Assessed continually. Toxicities evaluated using NCI CTCAE criteria v.5.0. **Serious Adverse Events (SAEs):** Standard reporting timelines; close monitoring for grade 3/4 hematologic and gastrointestinal toxicities. **Data Monitoring Committee (DMC):** External committee for independent safety data review during the Phase 2 to Phase 3 transition and dose-optimization phases.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints; Safety Set for AE/SAE evaluations.

Sample Size Justification: ~42 patients in Phase 2 to determine the recommended Phase 3 dose, expanding by ~188 patients in Phase 3 (Total targeted enrollment: 230). **Handling of Missing Data:** Adjustments for baseline imbalances utilizing a Cox proportional hazards model, incorporating variables such as ECOG status, LDH, RBC transfusions, Hgb, and cytogenetics.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring for data integrity, protocol adherence, and centralized pathology and cytogenetics review. **Audit Trail:** Independent central review of MDS diagnosis, WHO classification, and IPSS-R risk status locked via eCRF prior to randomization.

11. Study Identifiers & Governance

Primary ID: CA055-026 **NCT Number:** NCT05469737 **Posting ID:** 72607804122006579 **Source Level:** 5 **Sponsor/Company:** Bristol Myers Squibb **Study Title:** ASTREON

12. Clinical Framework (MDS Specific)

Disease Areas / Preferred UMLS Name: Myelodysplastic Syndromes
Name / Preferred Name: Myelodysplastic Syndrome (MYELODYSPLASTIC SYNDROME) **Semantic Type:** Neoplastic Process **Definition:** A clonal hematopoietic disorder characterized by dysplasia and ineffective hematopoiesis in one or more of the hematopoietic cell lines. The dysplasia may be accompanied by an increase in myeloblasts, but the number is less than 20%, which, according to the WHO guidelines, is the requisite threshold for the diagnosis of acute myeloid leukemia. It may occur de novo or as a result of exposure to alkylating agents and/or radiotherapy. (WHO, 2001) **MeSH Code:** D009190 **HPO Code:** HP:0002863 **SNOMED ID:** 109995007 **Diagnosis Threshold:** IPSS-R score between 1.5 and 4.5 with ≥ 1 cytopenia **Medical Methodology:** Hypomethylating agent therapy + Best Supportive Care

13. Study Procedures & Methodology

Management Pathway: Standard hematologic oncology care for MDS
Education Protocol (HCP): Investigator training on IWG 2006 response criteria and adverse event management **Education Protocol (Patient):** Oral dosing compliance instructions (14 days on, 14 days off) and cytopenia symptom tracking **Quality Audit Mechanism:**

Independent central pathology and cytogenetics review prior to randomization

14. Observation & Follow-Up Schedule

Total Duration: 6 cycles (approx. 24 weeks) for primary endpoint, followed by an optional extension phase or follow-up for OS and AML progression **Onsite Visit Cadence:** Day 1 of every 28-day cycle **Remote Monitoring Frequency:** As needed for AE management and RBC/Platelet tracking **Checklist Review Interval:** End of each 28-day cycle

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					